

Desigualdad de Chebyshev

Proposición 1 (Desigualdad de Chebyshev). Sea X una v.a. y $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ medible y no negativa, definimos $\forall A \in \mathcal{B}(\mathbb{R}) : i_A := \inf_{t \in A} \varphi(t)$

$$\implies i_A \cdot \mathbb{P}(X \in A) \leq \mathbb{E} [\mathbb{1}_{\{X \in A\}} \cdot \varphi(X)] .$$

Demostración: Sea $A \in \mathcal{B}(\mathbb{R})$, entonces $\forall \omega \in \Omega : X(\omega) \in A : i_A \leq \varphi(X(\omega))$

$$\implies i_A \cdot \mathbb{1}_{\{X \in A\}}(\omega) \leq \mathbb{1}_{\{X \in A\}}(\omega) \cdot \varphi(X(\omega))$$

Integrando, obtenemos que $i_A \cdot \mathbb{P}(X \in A) \leq \mathbb{E} [i_A \cdot \mathbb{1}_{\{X \in A\}}] \leq \mathbb{E} [\mathbb{1}_{\{X \in A\}} \cdot \varphi(X)]$. ■

Referenciado en

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